JIVAJ BRAR

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Relevant Experience

When2Surf — October 2025 - Present
Founding Software Engineer — Santa Cruz, CA

- Built solo across three codebases — Node.js/Express + MongoDB backend, React/TypeScript web app (Vercel), and native SwiftUI iOS app – sharing a production API on AWS with MongoDB Atlas; live in beta with 10+ users.
- Engineered a preference-matching engine evaluating live ocean forecasts (wave height, swell, wind, tide, water temp) against per-user tiered preferences with circular direction tolerance and per-spot overrides.
- Built a background scheduler that evaluates forecast windows, ranks candidate spots, deduplicates bookings, and auto-inserts Google Calendar events via OAuth 2.0 – core of a co-invented patent-pending scheduling system.

Cauris Finance — January 2025 - March 2025
Front-end Software Engineer Intern — San Francisco, CA

- Rebuilt an internal credit-risk chatbot into a fully-featured, cross-platform production React dashboard — responsive across mobile and desktop - consuming existing REST APIs for real-time financial data reporting and visualization.
- Designed and implemented responsive UI architecture with dynamic state management; debugged frontend-backend API interactions to ensure accurate data rendering across the dashboard.

Projects

RFVerify *RF Test Automation Framework* | Github

- Built a modular RF test framework that synthesizes configurable signals and injects real-world impairments (noise, distortion, interference) across instrument abstraction layers for a signal generator, oscilloscope, and spectrum analyzer.
- Implemented time-domain analysis (amplitude, RMS, noise floor, transients) and a full FFT pipeline (fftfreq bin mapping, DC removal, argmax peak detection) for harmonic classification and interference identification; validated with pytest.
- Designed an automated pass/fail engine (RFTestRunner) for frequency accuracy, harmonic count, and interference; prototyped an OpenTAP/C# test step packaging the same logic as a Verdict-based step in Keysight's PathWave environment, with a PyVISA-ready instrument control layer.

FraudSense *Fraud Detection System* | Github

- Built a backend service achieving 233 req/s and 6.33 ms p95 latency using FastAPI, Redis, PostgreSQL, and Docker; engineered ML features with scikit-learn, pandas, and NumPy, achieving ROC-AUC 0.941 on 250K transactions.

Papermaker *Algorithmic Trading Bot*

- Built an end-to-end trading pipeline in Python — scraped Reddit and Twitter/X via their APIs, ran NLP-based sentiment classification to generate buy/sell signals, and executed automated trades via QuantConnect.
- Outpaced the market by an average of 11% per quarter, achieving 28% annualized return across automated backtesting and live simulation

Inlyne *Real-Time Collaborative Documentation Platform* | Github

- Built a distributed real-time collaboration system using WebSocket/STOMP between a React frontend and Spring Boot backend, enabling concurrent multi-user document synchronization

SKILLS

Languages: Python, C/C++, C# (.NET), Java, JavaScript, TypeScript, Swift
Test Automation: OpenTAP, PathWave Test Automation, pytest, PyVISA (GPIB/USB/LAN instrument control), pass/fail test design, instrument abstraction
RF & Signal: Signal synthesis, FFT spectrum analysis, harmonic classification, noise/distortion injection, time-domain analysis
ML & Data: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, multi-agent systems, OpenAI/Gemini APIs
Backend & Infra: FastAPI, Spring Boot, Node.js/Express, REST APIs, WebSockets, AWS (EC2, ECS, Lambda, S3), Docker, Kubernetes, CI/CD, PostgreSQL, Redis, MongoDB

Education

University of California, Santa Cruz — September 2022 - June 2026
BS Computer Science | BA Business Management Economics — Santa Cruz, CA

- Relevant Coursework: Machine Learning, Applied ML, Artificial Intelligence, Computational Models, Computer Systems, Assembly Language, Data Structures and Algorithms, Distributed Systems, Natural Language Processing
- Placed top 5 of 102 students in classification accuracy in Applied ML course image classification competition using ResNet50 with PyTorch, TensorFlow, scikit-learn, pandas, and NumPy.